# th National High School Physics Competition: Final Solutions

1.

. Method 1: intercept at on , time .

Let be the - distance and the distance from to (Fig. ). Then

and Fig.

Solving gives

In , interception requires real roots, giving the necessary condition

Method 2: let be the angle between and (Fig. ). The sine rule gives

Using gives

gives the necessary condition

. Let be the least interception distance from . From ,

If the receiver is closer to than , he gets the ball first. The condition for avoiding interception is

From ,

By , if a receiver on , distance from , satisfies this inequality, the defender initially at cannot intercept the ball.

. Method 1: before the defender at reaches on , distance from ,